

Student Assignment Brief

This document is intended for Coventry University Group students for their own use in completing their assessed work for this module. It must not be passed to third parties or posted on any website. If you require this document in an alternative format, please contact your tutor.

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The work you submit for this assignment must be your own independent work, or in the case of a group assignment your own groups' work. More information is available in the '[Assignment Task](#)' section of this assignment brief.

Assignment Information

Module Name: Securing Cloud Systems

Module Code: 503AZ

Assignment Title: Group Security Project

Assignment Due:

- Group Evidence Submission – TBC
- Group Presentations – TBC
- Individual Reflection – Friday Week 6 @ 6PM

Assignment Credit: 30 Credits

Word Count (or equivalent):

- Group Presentation: 15 Mins (+/- 10%)

Assignment Type:

Percentage Grade (Applied Core Assessment). You will be provided with an overall grade between 0% and 100%. To pass the assignment you must achieve a grade of 40% or above.

Assignment Task

Scenario: CivicNexus Innovations

CivicNexus Innovations is a forward-thinking technology firm that partners with city governments to build smarter, more responsive urban environments. They have just launched a major initiative for a large metropolitan area: a centralized, cloud-based Smart City Operations Platform. This platform integrates data from across the city to streamline services and improve residents' quality of life. To spearhead this project, the company has appointed a new Head of Urban Technology. However, the platform was deployed under a tight deadline to meet contractual obligations, and the team is new to cloud infrastructure, resulting in a potentially insecure initial setup.

The infrastructure encompasses the following components:

- **Cloud Compute Resources:** CivicNexus Innovations utilizes a cloud network to support its diverse operational needs and specialised staff. The compute resources include:
 - **Elastic Web Servers:** Scalable web servers (such as Apache or Nginx) host public-facing portals displaying real-time traffic data and public transport schedules, as well as internal dashboards for city operators.
 - **Core Services Database Servers:** Highly available databases are deployed to manage critical city services, such as smart traffic light control systems and automated waste management dispatch.
 - **Virtual Desktop Infrastructure (VDI):** The firm uses virtual desktops hosted on cloud instances to provide urban planners and data scientists with secure, remote access to specialised geospatial and traffic simulation software.
- **Smart City Data Management:** CivicNexus has opted for a non-relational document database to manage the high volume and variety of data generated by the city's infrastructure with greater scalability. The document database stores the following information:
 - **Urban & Environmental Data:** The platform ingests and stores comprehensive records from across the city's infrastructure, including data from traffic flow monitors, air quality readings, waste collection statuses, and metadata from public safety cameras.
 - **Personnel Data:** CivicNexus maintains detailed records of city employees and contractors with access to the system, including names, job roles (e.g., Urban Planner, Field Technician), contact details, and security clearance levels.
 - **System & Operational Logs:** The database contains vital logs from automated systems, such as traffic light control sequences, public transport dispatch records, and maintenance schedules for critical infrastructure.

You will be given access to a baseline cloud infrastructure for CivicNexus that is deliberately and poorly configured. Your task is to work as a team to analyse, secure, and document the environment.

Your Group Task

Your group's mission is to design and partially implement a cohesive Cloud Security Framework. After a group-level risk analysis, your team will divide ownership of the infrastructure's assets (e.g., web server, database server). Each student will then be responsible for securing their assigned asset across four distinct layers: Network, Instance, Server, and Data. **Please email your module tutors what elements of the infrastructure each student will be analysing and securing**

The project follows this clear, three-step process:

1. **Risk Analysis:** Your group will first conduct an Asset-Threat-Vulnerability (ATV) analysis of the entire infrastructure.
2. **Propose & Implement Layered Controls:** Each student will implement a set of controls for their assigned asset, covering all four security elements.
3. **Create Security Framework:** Your group will integrate these multi-layered controls into a single, cohesive framework structured under ISO 27001 headings.

The overall complexity of the provided infrastructure will be scaled proportionally to your group size. This adjustment ensures that the individual responsibilities and workload for each student remain consistent, regardless of whether your group has two or three members

Whilst you are working as group you will be marked individually on the quality of your work. Please ensure that you label with your initials who completed what part of the work.

Phase 1: Group Project Development (50%)

Step 1: Group Asset-Threat-Vulnerability (ATV) Analysis (15%)

As a group, conduct an ATV analysis to identify CivicNexus 's key **Assets** (e.g., web server, database), **Threats** (e.g., ransomware, data breach), and **Vulnerabilities** (e.g., open firewall ports, weak passwords). Following this analysis, your group will assign one major asset/server to each member.

- **Deliverable:** A collaborative ATV Risk Analysis Report, including the assignment of assets to each team member.
- **Deadline:** TBC

Step 2: Implement Layered Controls (Individual Work) (20%)

This is the core practical task. For the specific asset you were assigned (e.g., the public-facing web server), you must individually design, implement, and document security controls across the following four elements:

- **Network Controls:** Configure specific firewall rules (Security Groups/NACLs) and network settings to enforce the principle of least privilege for your asset.
- **Instance (VM) Controls:** Secure the virtual machine instance itself, for example, by managing access keys, securing metadata endpoints, or choosing a hardened OS image.
- **Server (OS-Level) Controls:** Harden the operating system of your server. This could include disabling non-essential services, applying patches, or configuring user access and permissions within the OS.
- **Data Controls:** Secure the data that resides on or passes through your asset. This could involve file-level permissions, configuring data encryption at-rest, or implementing a backup plan for that specific data.
- **Deliverable:** Each student submits their own evidence in the form of a updated Statement of Applicability and Configuration Document, clearly detailing the implementation of controls for **all four elements** for their assigned asset (using annotated screenshots, scripts, AWS CLI exports etc.).
- **Deadline:**TBC

Templates will be provided on Aula

Step 3: Create Security Framework (Group Work) (15%)

In this step, your group will integrate the individual, multi-layered control sets into one comprehensive Cloud Security Framework document. You must ensure the controls do not conflict and work together to protect the entire system. This final report **must be structured using relevant headings from the ISO 27001 and the using the Annex A controls**, mapping your implemented controls to the appropriate domains (e.g., A.5, A.8, A.12). *Templates will be provided on Aula.*

- **Deliverable:** A single, co-authored Security Framework document (.docx) uploaded to Turnitin.
- **Deadline:** TBC

Phase 2: Group Presentation (30% of Module Mark)

In a **15-minute group presentation**, your group will present your completed Cloud Security Framework.

Your presentation must feature:

1. A summary of the key findings from your group's **ATV analysis**.
 2. Each member presenting the **four-layer security controls** they implemented for their specific asset.
 3. A final summary explaining how the secured assets integrate into a cohesive, ISO 27001-aligned framework.
- **Presentation Dates: Wednesday, Thursday, or Friday of Week 6.**

Phase 3: Individual Reflective Report (20% of Module Mark)

Following your presentation, you will individually submit a **750-word reflective report**. Your reflection must now consider the full stack of controls you implemented inline with Security and Cloud best practice and standards.

- **Report Deadline: Friday Week 6 @6PM.**

Guidance on AI and Academic Integrity

- **Category: Amber – AI Used to assist**

You may use generative AI tools to assist with brainstorming, refining grammar, structure and spelling in your written work, and debugging code. While responsible use of AI as a productivity tool is encouraged, **you are not permitted to use AI to generate functional code for your project**. Your tutors will demonstrate the required constructs, and generating code will likely cause you to miss core learning outcomes.

If you use AI to help debug an error, you must document it directly in your code comments as follows:

1. State the prompt you used.
2. Briefly explain the original problem.
3. In your own words, explain how the AI's suggestion fixed the issue and how the corrected code now works.

Authorship and Version Control Requirement: While AI tools can be used to assist, all submitted work must ultimately be your own. In accordance with the University's Academic Regulations, you are required to maintain a complete and regular commit history for both your source code and your technical presentation using a version-controlled repository (Git (or Similar) for your source code and University OneDrive for your presentations and diagrams).

Version history is the primary method of demonstrating the progression of your work and proving your authorship. In the event of an academic integrity query, your commit history is a formal part of the evidence base and may be called upon for review. Failure to provide a credible and consistent version of history may be treated as a breach of academic regulations. Submitting work that you cannot evidence as your own is a serious academic offence.

If using Artificial Intelligence (AI) tools in the development of your assignment, **you must reference which tools you have used and for what purposes** you have used them. This information must be acknowledged in your final submission before any references or bibliography which AI tools have been used and a brief explanation of how (see an example on **Academic Integrity** page on AULA). Students should save copies of their drafts as work progresses.

The below table should be included at the bottom of your assessment before the reference list if you have used a LLM in accordance with the above guidance:

AI Use Category	Tool	How AI Tool was used in this assignment
e.g. Inspiring (all categories listed here under the Example Uses of AI dropdown)	e.g. ChatGPT-3.5	Key word search on topics related to learning outcomes

Submission Instructions:

- TBC – ATV Analysis (.xlsx) uploaded to Turnitin via Aula (Group)
- TBC – Security Controls Zip File uploaded to HandIn via Aula (Individual)
- TBC – Security Framework (.docx) uploaded to Turnitin via Aula (Group)
- TBC (Prior to Live Presentations) - Group PowerPoint Presentation on Aula using Turnitin. (Group)
- Presentations (15 Mins) will take place Wednesday and Thursday of Week 6 times slots will be communicated on Aula.
- Friday Week 6 @ 6PM – Individual Reflection (750 Words) Uploaded to Turnitin via Aula

Marking and Feedback

How will my assignment be marked?

Your assignment will be marked by the module team

How will I receive my grades and feedback?

Provisional marks will be released once internally moderated.

Feedback will be provided by the module team alongside grades release; you will be able to access your feedback with through Aula or Email.

Your provisional marks and feedback should be available within 2 weeks (10 working days).

What will I be marked against?

Details of the marking criteria for this task can be found at the [bottom of this assignment brief](#).

Assessed Module Learning Outcomes

The Learning Outcomes for this module align to the [marking criteria](#) which can be found at the end of this brief.

Ensure you understand the marking criteria to ensure successful achievement of the assessment task. The following module learning outcomes are assessed in this task:

1. Appraise and evaluate the primary security principles of Confidentiality, Integrity and Availability, the security risks facing organisations and the consequences of security failures.
2. Describe and demonstrate an understanding of the development of security threats, legislation and regulations and security measures which can meet national and international standards.
3. Develop a framework and methodology to appraise and evaluate various security threats in a business scenario, perform risk analysis and recommend risk mitigation techniques.
4. Determine appropriate failover and redundancy approaches for cloud systems.
5. Demonstrate how security weaknesses occur and how these are exploited.
6. Demonstrate how to secure Cloud assets from attacks using various security measures.

Assignment Support and Academic Integrity

If you have any questions about this assignment please see the [Student Guidance on Coursework](#) for more information.

Spelling, Punctuation, and Grammar:

You are expected to use effective, accurate, and appropriate language within this assessment task.

Academic Integrity:

The work you submit must be your own, or in the case of groupwork, that of your group. All sources of information need to be acknowledged and attributed; therefore, you must provide references for all sources of information and acknowledge any tools used in the production of your work. We use detection software and make routine checks for evidence of academic misconduct.

It is your responsibility to keep a record of how your thinking has developed as you progress through to submission. Appropriate evidence could include: version controlled documents, developmental sketchbooks, or journals. This evidence can be called upon if we suspect academic misconduct.

If using Artificial Intelligence (AI) tools in the development of your assignment, you must reference which tools you have used and for what purposes you have used them. This information must be acknowledged in your final submission.

Definitions of academic misconduct, including plagiarism, self-plagiarism, and collusion can be found [on the Student Portal](#). All cases of suspected academic misconduct are referred for investigation, the outcomes of which can have profound consequences to your studies. For more information on academic integrity please visit the [Academic and Research Integrity](#) section of the Student Portal.

Support for Students with Disabilities or Additional Needs:

If you have a disability, long-term health condition, specific learning difference, mental health diagnosis or symptoms and have discussed your support needs with health and wellbeing you may be able to access support that will help with your studies.

If you feel you may benefit from additional support, but have not disclosed a disability to the University, or have disclosed but are yet to discuss your support needs it is important to let us know so we can provide the right support for your circumstances. Visit [the Student Portal](#) to find out more.

Unable to Submit on Time?

The University wants you to do your best. However, we know that sometimes events happen which mean that you cannot submit your assessment by the deadline or sit a scheduled exam. If you think this might be the case, guidance on understanding what counts as an extenuating circumstance, and how to apply is [available on the Student Portal](#).

Administration of Assessment

Assessment Author: Daniel Flood

Assessment Author Email: ac2081@coventry.ac.uk

Assignment Category: Portfolio (Group)

Attempt Type: Standard

Component Code: CW1

Assessment Marking Criteria

Mark band	Outcome	Guidelines
80-100%	Meets learning outcomes	Your work shows a masterful grasp of cloud security. The ATV analysis is exceptionally detailed, identifying subtle and complex risks. The implemented controls for the Network, Instance, Server, and Data layers are not just functional but elegant, efficient, and demonstrate a proactive, defence-in-depth strategy. Your final ISO 27001 framework is expertly structured, showing a deep understanding of the standard. Evidence of exceptional research into advanced threats relevant to a company like CivicNexus . The reflection shows a superior critical awareness of the CIA Triad in relation to your chosen controls. Demonstrates creative flair and originality, perhaps by implementing a sophisticated logging solution or a non-obvious, highly effective network segmentation strategy.
70-79%		An excellent and comprehensive security solution. The ATV analysis is thorough and accurate, and the implemented controls across the four layers are highly effective and well-documented. The group's ISO 27001 framework is logical, well-justified, and demonstrates a professional approach to security governance. Excellent ability to apply cloud provider documentation and security best practices to the CivicNexus scenario. The presentation and reflection provide a consistent, coherent, and substantiated argument for your security decisions. Work is completed with a high degree of accuracy, proficiency, and autonomy.
60-69%		A very strong submission that successfully secures the infrastructure. The ATV analysis correctly identifies the most significant risks. The security controls for the Network, Instance, Server, and Data layers are correctly implemented and functional. The final ISO 27001 framework is well-structured with clear mapping of your controls to the standard. Work demonstrates a strong understanding of core cloud security theories. The reflection shows a clear critical analysis of your component's effectiveness against the CIA Triad. The presentation is clear, and arguments are well-supported. Work is completed with accuracy and proficiency.
50-59%		A good and complete submission that meets the core requirements of the task. The ATV analysis is present and identifies the main risks, though it may lack some detail. All four control layers are addressed, but may contain some minor misconfigurations or inefficiencies. The ISO 27001 framework is logical but the justification for certain decisions may be inconsistent.

2:2		Demonstrates a good understanding of the concepts but with some gaps in critical analysis. The reflection demonstrates a logical argument but may not fully explore the nuances of Confidentiality, Integrity, and Availability. The work is mostly accurate and proficient
40-49%		Meets the basic learning outcomes. The ATV analysis is attempted but may be incomplete or miss obvious risks. The implementation of controls is basic, and one or more of the four layers (Network, Instance, Server, Data) may be missing or contain significant errors. The final ISO 27001 framework may be incomplete or poorly structured.
3 rd Class		The submission shows a basic understanding of how to secure a cloud environment but is sufficient to pass. The work may be incomplete, with some errors and irrelevant material present. The reflection is likely to be more descriptive than analytical.
0-39%	Fails to achieve learning outcomes	The learning outcomes have not been met. There is a clear failure to secure the CivicNexus infrastructure. Key deliverables like the ATV analysis or the ISO 27001 framework are absent or fundamentally flawed. The implemented controls are either missing or completely ineffective, leaving the system critically vulnerable.
Fail		Demonstrates very little understanding of the security concepts. The reflection is absent or shows no understanding of the task. Fundamental errors are present, and there is little to no evidence of research, problem-solving, or application of the provided learning resources.